

MUHAMMED FASIL PV

Software Architect | AI & IoT Developer | MVVM & Automation Expert

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Professional Summary

Innovative **Software Architect** with 13+ years of experience designing and deploying scalable, intelligent enterprise systems. Specialized in **AI-driven automation**, **IoT ecosystems**, **MVVM architectures**, and real-time retail infrastructures spanning cloud and edge. Delivered production systems across ERP, QMS, POS, and IoT automation with strong DevOps practices. Expert in **Flutter**, **Node.js**, **Python**, **PHP**, and **WPF (.NET)**. Built AI agents, chat automation (WhatsApp/Telegram), and secure retail networks using MikroTik and Dockerized microservices. Proven leadership in architecture planning, full-stack development, and CI/CD.

Core Competencies

Software Architecture and MVVM for **Flutter**, **Node.js**, and **WPF**

AI Agents and LLM Integration: OpenAI, LangChain, Qwen3, local AI

IoT Development: ESP32, Raspberry Pi, Arduino, ESL, MQTT

Chat and Workflow Automation: WhatsApp Business API, Telegram Bot API

Networking and Infrastructure: MikroTik RouterOS, WireGuard, NAT/DHCP, VPN, MQTT

DevOps and Cloud: Docker Compose, GitHub Actions, PM2, Nginx, GCP, AWS, Azure

Database Design and Optimization: MySQL indexing, caching, profiling, replication; hybrid sync

Security and Authentication: JWT, OAuth2, HTTPS, SSL, access policies

Professional Experience

Software Architect — Idol Technology LLC (Dubai, UAE)

Aug 2024 – Present

Architected **IdolMEA**, a next-generation **ERP** integrating **POS**, **QMS**, **IoT automation**, and **AI vision** modules.

Built **IdolQueue (i-QMS)** in **Flutter (MVVM)** and **Node.js** with **Hive** offline sync, **MQTT** event control, and multi-branch monitoring.

Developed **WhatsApp Business** and **Telegram** automation for queue alerts, AI responses, and service updates.

Deployed **IdolVPN mesh** using **MikroTik WireGuard** enabling secure multi-branch communication and API routing.

Integrated **AI theft detection (YOLO)** in self-checkout kiosks; automated alerts with **LED/GPIO** via **AIDL**.

Designed **LampAidlInterface (Kotlin + Flutter IPC)** for Android retail LED device control.

Created **hybrid cloud sync engine (MySQL ↔ Hive)** for offline-first analytics and reporting.

Senior Software Architect — Screl Info Pvt Ltd (India / UAE Branch)

Apr 2019 – Jan 2024

Designed and delivered **SincA ERP**: modular hybrid architecture (**Flutter + Node.js + AI backend in Python**).

Implemented **AI dashboards**, **OCR parsing**, and prediction engines using **TensorFlow** and **OpenAI APIs**.

Built **WhatsApp** and **Telegram** automation bots for notifications, invoice tracking, and support chat.

Deployed **Dockerized local servers** synchronized with **GCP APIs**; implemented **CI/CD pipelines**.

Designed **AI queue prediction**, **dynamic pricing logic**, and **custom module generators**.

Mentored multi-location dev teams on **MVVM**, **modular API design**, and **DevOps workflows**.

Software Developer — Sincap Info Pvt Ltd (Calicut, India)

Mar 2014 – Mar 2019

Developed **ERP**, **accounting**, and **billing systems** using **WPF (C#)**, **PHP**, and **SQL Server**.

Built admin and reporting portals with **PHP**, **RESTful JSON endpoints**, and **role-based dashboards**.

Improved **UI/UX performance**, introduced **caching**, and **modularized** critical components.

Provided enterprise client support (50+ businesses), **database optimization**, and **automated reporting**.

Major Projects

IdolMEA ERP: Centralized ERP for retail/service—POS, booking, QMS, IoT device control, AI reporting.

IdolQueue (i-QMS): Smart queue management—Flutter + Node.js with MQTT-based automation.

Self-Checkout AI Kiosk: Android kiosk integrating YOLO detection and LED/GPIO control.

IdolVPN Network: Multi-branch VPN mesh using MikroTik WireGuard for secure operations.

BookzNow: Flutter MVVM service booking platform for salons and clinics.

Vlogfied App: Video-based restaurant discovery with AI content classification and tagging.

ESL Tools (OEPL / Solum): Firmware flashing, tag testing jigs, MQTT-based ESL control automation.

Local AI Agents: On-device OpenAI + Qwen3 agents for store and ERP automation.

Technical Skills

Languages: Dart, JavaScript, TypeScript, C#, Python, PHP

Frameworks: Flutter, Node.js, Express.js, .NET (WPF), TensorFlow, PyTorch

AI and LLM: OpenAI, LangChain, Qwen3, Local AI Agents, Whisper, Stable Diffusion

Automation: WhatsApp Business API, Telegram Bot API, Python schedulers/cron

IoT and Hardware: ESP32, Arduino, Raspberry Pi, Solum ESL, OEPL, MQTT

UI and UX: Figma, Material 3, TailwindCSS, Responsive/Adaptive design

Networking: MikroTik RouterOS, WireGuard, MQTT, NAT, VPN

Databases: MySQL, Hive, ObjectBox, Firebase, MongoDB

Cloud and DevOps: Docker, Docker Compose, GitHub Actions, PM2, Nginx, GCP, AWS, Azure

Security: JWT, OAuth2, HTTPS, SSL Certificates, VPN management

Education

Diploma in Computer Engineering — JDT Islam Polytechnic, Calicut, Kerala, India

June 2010 – March 2013

Additional Information

Nationality: Indian | Visa Status: Company Visa

Languages: English, Malayalam, Hindi

Portfolio: faztrick.com